Actuarial Exam P — Practice Exam 1

Time Allowed: 3 hours (180 minutes)
Total Questions: 30
Format: Multiple Choice (5 options per question)

Question 1 (Topic 1: General Probability)

A survey of an insurance company's policyholders regarding their sports viewing habits over the past year revealed the following data:

- 28% watched gymnastics
- 29% watched baseball
- 19% watched soccer
- 14% watched gymnastics and baseball
- 12% watched baseball and soccer
- 10% watched gymnastics and soccer
- 8% watched all three sports

Calculate the percentage of policyholders who watched **none** of the three sports during the past year.

(A) 24%
(B) 36%
(C) 48%
(D) 52%
(E) 64%

Question 2 (Topic 1: General Probability)

A department consists of 6 senior actuaries and 4 junior actuaries. A special project committee of 4 actuaries is to be chosen at random without replacement from this department.

Calculate the probability that the committee contains **at least 2 junior actuaries**.

(A) 0.3810
(B) 0.4524
(C) 0.5476
(D) 0.6190
(E) 0.7143

Question 3 (Topic 1: General Probability)

A medical diagnostic test for a rare disease has a sensitivity of 95% (probability of testing positive given that the disease is present) and a specificity of 90% (probability of testing negative given that the disease is not present). In the general population, 2% of individuals have the disease.

Calculate the conditional probability that a randomly selected individual has the disease given that they test positive.

(A) 0.1624
(B) 0.1900
(C) 0.8376
(D) 0.9000
(E) 0.9500

Question 4 (Topic 1: General Probability)

An auto insurer classifies drivers into three risk categories: Low Risk (30% of policyholders), Medium Risk (50% of policyholders), and High Risk (20% of policyholders). The annual probability of filing a claim for each category is:

- Low Risk: 0.01
- Medium Risk: 0.04
- High Risk: 0.10

Given that a randomly chosen policyholder files a claim during the year, calculate the probability that the driver is High Risk.

(A) 0.2000
(B) 0.3125
(C) 0.4651
(D) 0.5349
(E) 0.6250

Question 5 (Topic 1: General Probability)

Let A and B be independent events such that $P(A) = 0.40$ and $P(B) = 0.50$.

Calculate $P(A^c \cap B^c)$.

(A) 0.10
(B) 0.20
(C) 0.30
(D) 0.40
(E) 0.70

Question 6 (Topic 1: General Probability)

An actuary knows that $P(A \cup B) = 0.70$ and $P(A) = 0.40$. Assuming that events A and B are **independent**, calculate $P(B)$.

(A) 0.30
(B) 0.40
(C) 0.50

(D) 0.60
(E) 0.75

Question 7 (Topic 1: General Probability)

A discrete random variable X has probability mass function:

$$p(x) = kx^2, \quad x \in \{1, 2, 3, 4\}$$

Calculate $P(X \geq 3)$.

(A) 0.3000
(B) 0.5333
(C) 0.7667
(D) 0.8333
(E) 0.9000

Question 8 (Topic 1: General Probability)

An actuary arranges 9 claim files on a shelf: 4 red files (Auto), 3 blue files (Property), and 2 green files (Liability). Files of the same color are indistinguishable from each other.

Calculate the total number of distinct linear arrangements possible for these 9 files.

(A) 288
(B) 504
(C) 1260
(D) 2520
(E) 362,880

Question 9 (Topic 2: Univariate Random Variables)

An actuary has discovered that policyholders are three times as likely to file two claims as to file four claims in a given year. The number of claims filed follows a Poisson distribution.

Calculate the variance of the number of claims filed.

(A) $\frac{1}{\sqrt{3}}$
(B) 1
(C) $\sqrt{2}$
(D) 2
(E) 4

Question 10 (Topic 2: Univariate Random Variables)

The number of sales calls an agent makes before securing their first policy sale follows a Geometric distribution on $\{1, 2, 3, \dots\}$. The probability that the agent requires more than 4 calls to secure the first sale

is 0.4096.

Calculate the expected number of calls needed to secure the first sale.

(A) 2.5
(B) 4.0
(C) 5.0
(D) 6.25
(E) 8.0

Question 11 (Topic 2: Univariate Random Variables)

An insurer sells 10 independent group life insurance policies. Each policy has a 0.20 probability of incurring a claim during the year.

Calculate the probability that **at least 2** policies incur a claim during the year.

(A) 0.2684
(B) 0.3758
(C) 0.6242
(D) 0.7316
(E) 0.8926

Question 12 (Topic 2: Univariate Random Variables)

The loss X sustained by a commercial business has probability density function:

$$f(x) = cx(2 - x), \quad 0 < x < 2$$

Calculate the median loss of X.

(A) 0.50
(B) 0.75
(C) 1.00
(D) 1.25
(E) 1.50

Question 13 (Topic 2: Univariate Random Variables)

Losses X under an auto insurance policy follow an Exponential distribution with mean 10. The policy covers losses subject to a deductible of 4.

Calculate the expected claim payment made by the insurer per loss.

(A) $10e^{-0.4}$
(B) $10e^{-2.5}$
(C) $6e^{-0.4}$
(D) $10(1 - e^{-0.4})$
(E) $4e^{-0.4}$

Question 14 (Topic 2: Univariate Random Variables)

Losses X follow a Continuous Uniform distribution on the interval $[0, 100]$. An insurance policy covers losses subject to an ordinary deductible of 20 and a maximum benefit limit of 60 (so the maximum payment is 60).

Calculate the expected claim payment under this policy.

(A) 24
(B) 30
(C) 36
(D) 40
(E) 48

Question 15 (Topic 2: Univariate Random Variables)

Losses X follow an Exponential distribution with mean 100. A policy has a maximum benefit limit of 50 with zero deductible (so the payment is $Y = \min(X, 50)$).

Calculate $E[Y]$.

(A) $100(1 - e^{-0.5})$
(B) $100e^{-0.5}$
(C) $50(1 - e^{-0.5})$
(D) $50e^{-0.5}$
(E) $100(1 - e^{-2})$

Question 16 (Topic 2: Univariate Random Variables)

Automobile loss amounts are modeled by a uniform distribution on $[0, 10000]$. The insurance company covers each loss subject to a deductible of 2000.

Calculate the expected claim payment **given that a payment is made** (per-claim expected payment).

(A) 2000
(B) 3200
(C) 4000
(D) 5000
(E) 6000

Question 17 (Topic 2: Univariate Random Variables)

A loss random variable X follows a Gamma distribution with shape parameter $\alpha = 3$ and scale parameter $\theta = 4$.

Calculate $E[X^2]$.

(A) 48
(B) 144
(C) 192
(D) 240
(E) 288

Question 18 (Topic 2: Univariate Random Variables)

Annual claim amounts on a commercial property policy follow a Normal distribution with mean 500 and standard deviation 100.

Calculate the probability that a claim exceeds 650.

(A) 0.0228
(B) 0.0668
(C) 0.1587
(D) 0.8413
(E) 0.9332

Question 19 (Topic 2: Univariate Random Variables)

Let X be uniformly distributed on the interval $(0, 2)$. Let $Y = X^2$.

Find the probability density function $f_Y(y)$ of Y for $0 < y < 4$.

(A) $\frac{1}{4}$
(B) $\frac{1}{2\sqrt{y}}$
(C) $\frac{1}{4\sqrt{y}}$
(D) $\frac{\sqrt{y}}{4}$
(E) $\frac{1}{8\sqrt{y}}$

Question 20 (Topic 2: Univariate Random Variables)

A continuous loss variable X has probability density function:

$$f(x) = 2x, \quad 0 < x < 1$$

Calculate the absolute difference between the mode and the mean of X.

(A) 0.1667
(B) 0.2500
(C) 0.3333
(D) 0.5000
(E) 0.6667

Question 21 (Topic 2: Univariate Random Variables)

Claim severity X follows an Exponential distribution with mean 20. An insurance policy pays 80% of the claim amount in excess of a deductible of 5.

Calculate the expected payment under this policy.

(A) 11.23
(B) 12.46
(C) 14.80
(D) 15.58
(E) 16.00

Question 22 (Topic 2: Univariate Random Variables)

Losses X follow an Exponential distribution with mean 200. An auto policy has an ordinary deductible of 100 and a maximum benefit limit of 400 (maximum claim payment of 400).

Calculate the expected claim payment made for a loss.

(A) 84.15
(B) 92.40
(C) 104.89
(D) 121.31
(E) 142.60

Question 23 (Topic 3: Multivariate Random Variables)

The joint probability density function of X and Y is given by:

$$f(x,y) = c(x+2y), \quad 0<x<1,\, 0<y<1$$

Calculate the constant c.

(A) $\frac{1}{3}$
(B) $\frac{1}{2}$
(C) $\frac{2}{3}$
(D) $\frac{3}{4}$
(E) 1

Question 24 (Topic 3: Multivariate Random Variables)

The continuous random variables X and Y have joint probability density function:

$$f(x,y) = 8xy, \quad 0<x<y<1$$

Find the marginal density function $f_X(x)$ for $0<x<1$.

(A) $4x$
(B) $4x(1-x^2)$
(C) $8x(1-x)$

(D) $2x(1-x^2)$
(E) $4x^3$

Question 25 (Topic 3: Multivariate Random Variables)

Using the joint density $f(x,y) = 8xy$ for $0 < x < y < 1$, calculate the conditional expectation $E[Y \mid X = 0.5]$.

(A) 0.5000
(B) 0.6667
(C) 0.7500
(D) 0.7778
(E) 0.8333

Question 26 (Topic 3: Multivariate Random Variables)

The annual number of claims N for a policyholder given $\Lambda = \lambda$ follows a Poisson distribution with mean λ. The parameter Λ itself is a continuous random variable uniformly distributed on $[1, 5]$.

Calculate $Var(N)$.

(A) 3.00
(B) 3.33
(C) 4.00
(D) 4.33
(E) 5.33

Question 27 (Topic 3: Multivariate Random Variables)

An actuary analyzes personal auto claims M and commercial auto claims N. You are given:

- $Var(M) = 1600$
- $Var(N) = 900$
- Correlation $\rho_{M,N} = 0.64$

Calculate $Var(M+N)$.

(A) 2500
(B) 3268
(C) 4036
(D) 4420
(E) 5036

Question 28 (Topic 3: Multivariate Random Variables)

A machine has 3 identical components operating independently. The lifetime X_i of component i is exponentially distributed with mean 2 years ($F(x) = 1 - e^{-x/2}, x > 0$).

Calculate the probability that the largest of the 3 component lifetimes is less than 3 years.

(A) 0.2231
(B) 0.3542
(C) 0.4690
(D) 0.7769
(E) 0.8842

Question 29 (Topic 3: Multivariate Random Variables)

Two independent systems have lifetimes X_1 and X_2 that are exponentially distributed with failure rates $\lambda_1 = 0.10$ and $\lambda_2 = 0.20$ per year, respectively.

Calculate the expected value of $W = \min(X_1, X_2)$.

(A) 2.50
(B) 3.33
(C) 5.00
(D) 6.67
(E) 15.00

Question 30 (Topic 3: Multivariate Random Variables)

An insurer has 100 independent health insurance policies. The claim amount for each policy has mean $\mu = 500$ and standard deviation $\sigma = 100$.

Using the Central Limit Theorem, calculate the approximate probability that the total aggregate claims for the 100 policies exceed 52,000.

(A) 0.0228
(B) 0.0668
(C) 0.1587
(D) 0.3085
(E) 0.4772

Answer Key & Solutions — Practice Exam 1

1. **(D) 52%**
   $P(A \cup B \cup C) = (28 + 29 + 19) - (14 + 12 + 10) + 8 = 76 - 36 + 8 = 48\%$.
   $P(\text{None}) = 100\% - 48\% = \mathbf{52\%}$.

2. **(C) 0.5476**
   Total sample space $\binom{10}{4} = 210$.
   Favorable: $\binom{4}{2}\binom{6}{2} + \binom{4}{3}\binom{6}{1} + \binom{4}{4}\binom{6}{0} = 6(15) + 4(6) + 1(1) = 90 + 24 + 1 = 115$.
   $P = \frac{115}{210} = \mathbf{0.5476}$.

3. **(A) 0.1624**
   $P(D \mid +) = \frac{P(+|D)P(D)}{P(+|D)P(D)+P(+|D^c)P(D^c)} = \frac{(0.95)(0.02)}{(0.95)(0.02)+(0.10)(0.98)} = \frac{0.019}{0.019+0.098} = \mathbf{0.1624}$.

4. **(C) 0.4651**
   $P(\text{Claim}) = 0.30(0.01) + 0.50(0.04) + 0.20(0.10) = 0.003 + 0.020 + 0.020 = 0.043$.
   $P(\text{High} \mid \text{Claim}) = \frac{0.20(0.10)}{0.043} = \frac{0.020}{0.043} = \mathbf{0.4651}$.

5. **(C) 0.30**
   By independence of complements: $P(A^c \cap B^c) = P(A^c)P(B^c) = (1-0.40)(1-0.50) = 0.60 \times 0.50 = \mathbf{0.30}$.

6. **(C) 0.50**
   $P(A \cup B) = P(A) + P(B) - P(A)P(B) \implies 0.70 = 0.40 + P(B)(1-0.40) \implies 0.30 = 0.60P(B) \implies P(B) = \mathbf{0.50}$.

7. **(D) 0.8333**
   $\sum p(x) = k(1^2 + 2^2 + 3^2 + 4^2) = 30k = 1 \implies k = 1/30$.
   $P(X \geq 3) = \frac{3^2+4^2}{30} = \frac{25}{30} = \mathbf{0.8333}$.

8. **(C) 1260**
   Arrangements with repeated items $= \frac{9!}{4!3!2!} = \frac{362880}{24 \times 6 \times 2} = \mathbf{1260}$.

9. **(D) 2**
   $P(N=2) = 3P(N=4) \implies \frac{e^{-\lambda}\lambda^2}{2!} = 3\frac{e^{-\lambda}\lambda^4}{4!} \implies \frac{\lambda^2}{2} = \frac{3\lambda^4}{24} \implies \lambda^2 = 4 \implies \lambda = 2$.
   For Poisson, $Var(N) = \lambda = \mathbf{2}$.

10. **(C) 5.0**
    $P(X > 4) = (1-p)^4 = 0.4096 \implies 1-p = 0.8 \implies p = 0.2$.
    $E[X] = \frac{1}{p} = \frac{1}{0.2} = \mathbf{5.0}$.

11. **(C) 0.6242**
    $P(X \geq 2) = 1 - P(X=0) - P(X=1) = 1 - (0.8)^{10} - 10(0.2)(0.8)^9 = 1 - 0.10737 - 0.26844 = \mathbf{0.6242}$.

12. **(C) 1.00**
    $\int_0^2 c(2x - x^2)dx = c[x^2 - x^3/3]_0^2 = \frac{4}{3}c = 1 \implies c = \frac{3}{4}$.
    The density $f(x) = \frac{3}{4}x(2-x)$ is symmetric about $x = 1$. Hence, median $= \mathbf{1.00}$.

13. **(A)** $10e^{-0.4}$
    $E[Y] = \int_4^\infty (x-4)\frac{1}{10}e^{-x/10}dx = 10e^{-4/10} = \mathbf{10e^{-0.4}}$.

14. **(B) 30**
    $E[Y] = \int_0^{60} S_Y(y)dy = \int_0^{60} P(X - 20 > y)dy = \int_0^{60} \frac{100-(20+y)}{100}dy = \frac{1}{100}\int_0^{60}(80-y)dy = \frac{1}{100}[80(60) - 1800] = \mathbf{30}$.

15. **(A)** $100(1 - e^{-0.5})$
    $E[X \wedge 50] = \int_0^{50} S(x)dx = \int_0^{50} e^{-x/100}dx = 100(1 - e^{-50/100}) = \mathbf{100(1 - e^{-0.5})}$.

16. **(C) 4000**
    Given $X > 2000$, $(X - 2000) \sim Uniform(0, 8000)$.
    $E[X - 2000 \mid X > 2000] = \frac{8000}{2} = \mathbf{4000}$.

17. **(C) 192**
$E[X] = \alpha\theta = 12$, $Var(X) = \alpha\theta^2 = 48$.
$E[X^2] = Var(X) + (E[X])^2 = 48 + 144 = \mathbf{192}$.

18. **(B) 0.0668**
$Z = \frac{650-500}{100} = 1.50$.
$P(Z > 1.50) = 1 - \Phi(1.50) = 1 - 0.9332 = \mathbf{0.0668}$.

19. **(C)** $\frac{1}{4\sqrt{y}}$
$F_Y(y) = P(X^2 \le y) = P(X \le \sqrt{y}) = \frac{\sqrt{y}}{2}$.
$f_Y(y) = \frac{d}{dy}\left(\frac{\sqrt{y}}{2}\right) = \mathbf{\frac{1}{4\sqrt{y}}}$.

20. **(C) 0.3333**
Mean $= \int_0^1 x(2x)dx = \frac{2}{3}$. Mode $= 1$ (since density increases linearly up to $x = 1$).
$|1 - 2/3| = \frac{1}{3} = \mathbf{0.3333}$.

21. **(B) 12.46**
$Y = 0.8\max(0, X - 5)$. Memoryless property: $E[X - 5 \mid X > 5] = 20$.
$P(X > 5) = e^{-5/20} = e^{-0.25} \approx 0.7788$.
$E[Y] = 0.8 \times 20 \times e^{-0.25} = 16e^{-0.25} \approx \mathbf{12.46}$.

22. **(C) 104.89**
$E[Y] = \int_{100}^{500} S(x)dx = \int_{100}^{500} e^{-x/200}dx = 200(e^{-0.5} - e^{-2.5}) = 200(0.60653 - 0.08208) = \mathbf{104.89}$.

23. **(C)** $\frac{2}{3}$
$\int_0^1\int_0^1 c(x+2y)dxdy = c\int_0^1(1/2 + 2y)dy = c(1/2+1) = 1.5c = 1 \implies c = \mathbf{\frac{2}{3}}$.

24. **(B)** $4x(1-x^2)$
$f_X(x) = \int_x^1 8xydy = [4xy^2]_x^1 = \mathbf{4x(1 - x^2)}$.

25. **(D) 0.7778**
$f_{Y|X}(y \mid x) = \frac{8xy}{4x(1-x^2)} = \frac{2y}{1-x^2}$ for $x < y < 1$.
For $x = 0.5$, $f(y \mid 0.5) = \frac{2y}{0.75} = \frac{8}{3}y$ for $0.5 < y < 1$.
$E[Y \mid X = 0.5] = \int_{0.5}^1 y\frac{8}{3}ydy = \frac{8}{3}\left[\frac{y^3}{3}\right]_{0.5}^1 = \frac{8}{9}(1 - 0.125) = \frac{7}{9} = \mathbf{0.7778}$.

26. **(D) 4.33**
$E[N \mid \Lambda] = \Lambda$, $Var(N \mid \Lambda) = \Lambda$.
For $\Lambda \sim Uniform(1, 5)$, $E[\Lambda] = 3$, $Var(\Lambda) = \frac{(5-1)^2}{12} = \frac{16}{12} = \frac{4}{3}$.
$Var(N) = E[Var(N \mid \Lambda)] + Var(E[N \mid \Lambda]) = 3 + \frac{4}{3} = \frac{13}{3} = \mathbf{4.33}$.

27. **(C) 4036**
$Cov(M, N) = 0.64\sqrt{1600}\sqrt{900} = 0.64(40)(30) = 768$.
$Var(M + N) = 1600 + 900 + 2(768) = \mathbf{4036}$.

28. **(C) 0.4690**
$F(3) = 1 - e^{-3/2} = 1 - 0.22313 = 0.77687$.
$P(\max(X_1, X_2, X_3) < 3) = [F(3)]^3 = (0.77687)^3 = \mathbf{0.4690}$.

29. **(B) 3.33**
    Minimum of independent exponentials is Exponential with rate $\lambda = \lambda_1 + \lambda_2 = 0.10 + 0.20 = 0.30$.
    $E[W] = \frac{1}{0.30} = \mathbf{3.33}$.

30. **(A) 0.0228**
    $E[S] = 100(500) = 50,000$. $Var(S) = 100(100^2) = 1,000,000 \implies \sigma_S = 1000$.
    $P(S > 52000) = P\left(Z > \frac{52000-50000}{1000}\right) = P(Z > 2.00) = 1 - 0.9772 = \mathbf{0.0228}$.